

ArchFramework

ArchFramework is a modular .NET framework for building event-driven applications with explicit runtime workflows, provider-neutral storage, identity, notifications, execution tracking, and reusable test fakes.

The repository is organized around small libraries. Each library has a focused README and a local wiring diagram that shows which services to instantiate or inject, which interfaces providers implement, and which methods are normally called.

Repository structure

```
src/           Framework libraries and provider implementations
  tests/       Unit tests for framework libraries
  samples/     Small runnable sample applications
```

Libraries

Leva.Framework.Core	Shared contracts, results, IDs, clocks, entries, snapshots, synchronization helpers
Leva.Framework.Engine	Event queue, dispatcher, state machine, routines, behaviors, boards, runtime log
Leva.Framework.Execution	Execution runner, execution board, entries, status, progress, cancellation
Leva.Framework.Fakes	Test doubles and shared result assertions
Leva.Framework.Storage	Provider-neutral repository and journal contracts
Leva.Framework.Storage.Memory	In-process storage provider
Leva.Framework.Storage.Files	Local JSON/file-system storage provider
Leva.Framework.Storage.Sqlite	Local SQLite storage provider
Leva.Framework.Identity	Principals, sessions, authentication, authorization, audit contracts
Leva.Framework.Identity.Memory	In-memory identity provider
Leva.Framework.Identity.Local	Local/offline credential provider
Leva.Framework.Identity.AspNet	ASP.NET Core identity integration
Leva.Framework.Notifications	Provider-neutral notification model, gateway, store, service
Leva.Framework.Notifications.Memory	In-memory notification provider
Leva.Framework.Notifications.SignalR	ASP.NET Core SignalR notification provider

Samples

Leva.Framework.Sample.CounterWorkflow	Core, Engine, Execution, and Fakes in a small workflow
Leva.Framework.Sample.TicketDesk	Storage providers, Identity.Memory, and Notifications.Memory in a ticketing flow
Leva.Framework.Sample.LiveDashboard	Notifications.SignalR in a minimal live web dashboard
Leva.Framework.Sample.LocalAccount	Identity.Local and Identity.AspNet in a minimal account host

Documentation convention

Each library contains:

```
README_*.md      Local explanation of the library
DIAGRAM_*.png    Wiring diagram for normal use
```

Diagram colors are consistent across libraries:

```
Green   Application or tests
Blue    Framework or provider implementation
Yellow  Interface or contract
White   Value, entry, payload, ID, or model
```

Function and method names are written in italics and include `()`. Arrow labels use a small shared vocabulary such as `uses`, `calls`, `creates`, `implements`, `stores`, `returns`, `sends`, and `maps`.

Build and test

From the repository root:

```
dotnet restore ArchFramework.slnx
dotnet build ArchFramework.slnx
dotnet test ArchFramework.slnx
```

Run one sample:

```
dotnet run --project samples/Leva.Framework.Sample.CounterWorkflow
```